

THE REAL SYSTEM

VIIRS active fire detections over 8 wildfires and 14 to 20 consecutive days. 18k observed fire days with 11 remote sensing drivers. Gridded meteorology.

observe

PERCEIVE

$$\hat{u}_t = m_t y_t + (1 - m_t) \Phi$$

frozen backbone, LoRA, decoder

fills what was not observed

60 percent hidden costs only

6.6 of 41 decision points

imputation lost in 0 of 9 runs

SIMULATE

$$u_{t+1} = G_\theta(u_t, a_t)$$

U-Net fronts, FNO smooth fields

learned gate, rollout and physics

0.3162 on the NDWS test split,

111 percent of the published

gate declines the spectral path

PLAN

$$\min_b E[\text{loss}] \text{ s.t. } \bar{b}_t \leq B$$

projected gradient, decision time

PID dual, probe, conformal margin

12.0 points over the operational

heuristic, paired t of 11.15

violations 7.3 percent down to 0

intervene

EXPLAIN

$$F(b) = \exp(-\text{KL}/K)$$

brief parsed to a distribution

split conformal certificate

coverage holds at 90 and 95

percent on 3 of 3 seeds

faithfulness claim withdrawn

STATE SYNC

replace the belief with what the satellite actually saw

8.4 times better at one day
8 of 8 fires, t of 15.9

MODEL ADAPTATION

one gradient step on the model from each observed day

raises day 2 by 21 percent
and day 3 by 30 percent

REALITY PROBE

roll the policy in the true system, feed that cost to the dual

conformal margin holds the stated rate at delta of 0.1

THE PATH THAT DOES NOT CLOSE

No observational record contains a fire where a firebreak was cut on our instruction. Intervention effects rest on a stated action model, and no change of hazard or dataset repairs that.

PATHS THAT CARRY REAL MEASUREMENT BACK INTO THE LOOP

Green paths are validated against real observations. The dashed red path is model based. That boundary is the honest scope of the twin claim.